Conformation and configuration of cyclic

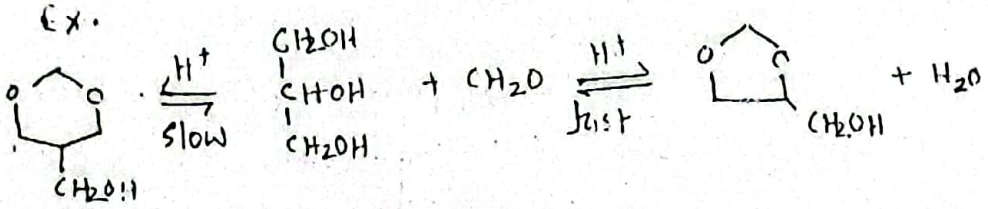
1. Ease of ring formation:

Ease of ring formation means that how easily a ring formed from its acyclic reactants. This facts we can explain by several factors involved in the ring formation like.

[1] As a function of ring size:

The easeness of ring formation mainly depend on the ring size. Some time we find to decrease probability of forming a ring with increasing the no. of ring constituents.

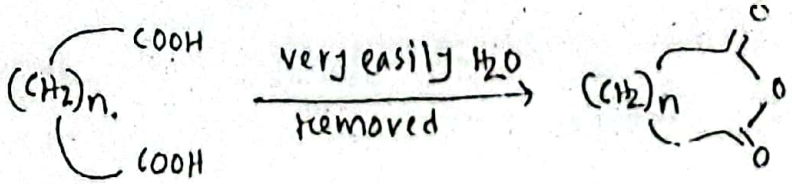
Ex.



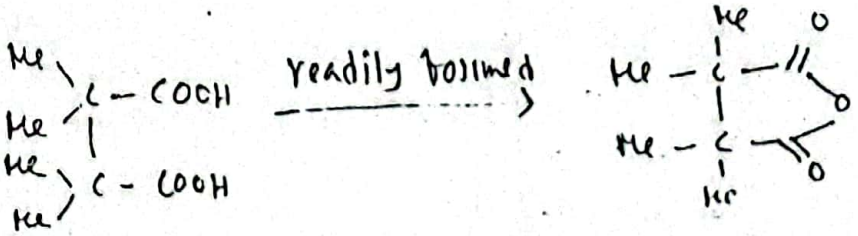
5 membered ring formation is more easier here.

[2] Effect of substituents:

Some substituent enhance the ring formation. As like in the formation of Lactone, cyclic anhydride relatively easier.



Ex.

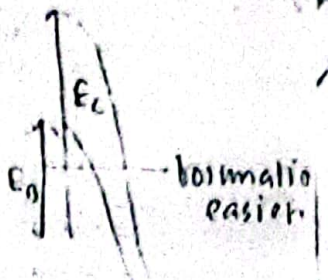


[3]. Nature of the ring: Introduction of any hetero atom in replacement of methylene group extremely slower the ring closure.

Ex. ring closure of Br(CH2)4NH2 is extremely slow.

Fig: From Kinetics, we can determine what which ring closure is easier.

The transition energy having lower energy is relatively easier to form.



$$C \rightleftharpoons A \rightleftharpoons B$$

ii) Stability of ring compounds:

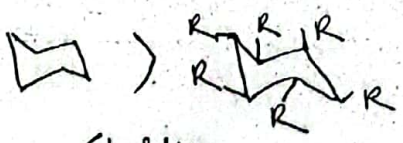
Stability of rings may differ on respect of various purpose as like following -

1. No. of ring members:

Ring may be 3 membered, 4-membered, 5 and 6 membered etc. The stability of cyclopropane is found lowest and Cyclohexane highest.

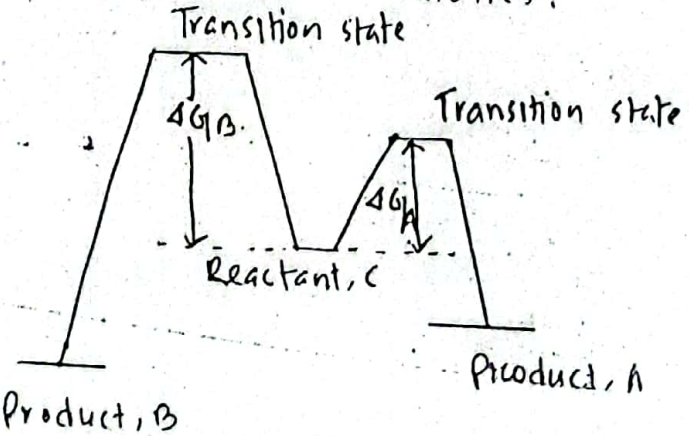
2. Nature of ring:

3. Substituent effect: Crowdness of substituents increase the steric interaction results less stable ring.



Stability

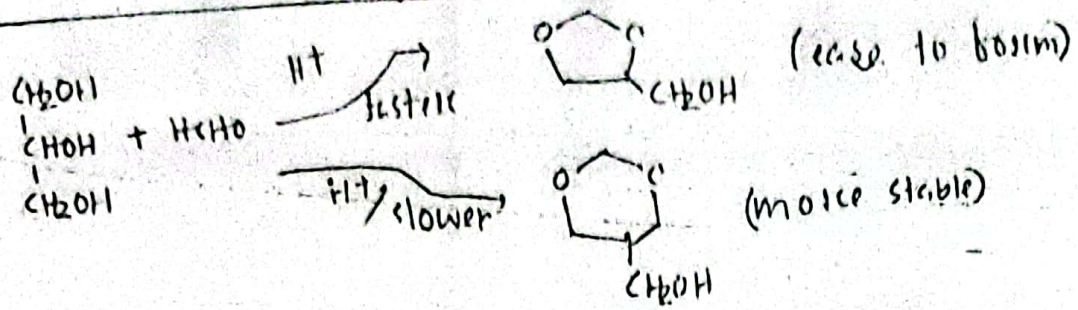
iii) Stability, Ease vs Kinetics:



Ease of ring formation is due to lower energetic tr state.

But, Product B is more st since lower energy.

2.



(ease to form)

(Based on the relative heat of combustion)

n (ring size)	Hc/n	Hc/n − 157.4	
3	166.6	9.2	} Maximum
4	163.95	6.55	
5	158.7	1.3	
6	157.4	0.0	minimum
7	158.3	0.9	
8	158.6	1.2	
9	158.8	1.4	rise to Max.
10	158.6	1.2	
11	158.4	1.0	
12	157.7	0.3	
13	157.8	0.4	
14	157.1	0.0	min.

The strain of the ring varies from minimum to maximum, again to minimum. Now, according to this concept we classify rings like following -

Small rings = 3,4 membered ⇒ High Hc, high strain.
Common rings = 5,6,7 " ⇒ Little strain
Medium ring = 8-11 " ⇒ High Hc.
Large ring = 12-others " ⇒ Relatively less Hc.

(dD/dt)

(i) Bayer strain theory:

Bayer postulated that any destiv deviation of bond angles from the normal tetrahedrical value would impose a condition of internal strain on the ring. This is called ang strain.

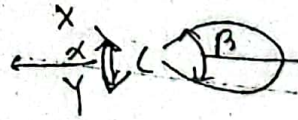
Compnd	Bond angle (θ)	Angle strain [½(109.28 − θ)]
↔	60°	24.44°
□	90°	9.44°
⬠	108°	0.44°
⬡	120°	−5.16° etc.

According to this theory cyclopropane has max. angle strain and ~~inc~~ decreases for cyclobutane. Minimum strain for cyclopro But the value indicates that cyclohexane should be relatively unstable which is practically not bound.

Thus, this theory accounts well for small rings but not valid for larger.

(ii) Ingold – Thorpe postulate:

The diminution of the internal angle in a small ring leads to "a spreading" apart of the external angle. This in turn relieves steric compression betwn substituents attached to one and the same carbon. thus favouring the ring form over the open chain form.

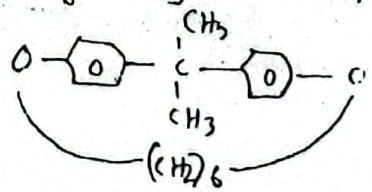


external angle → internal angle

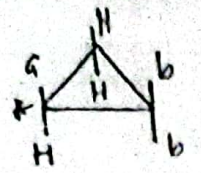
$\alpha > 109^\circ 28'$

$\beta < 109^\circ 28'$.

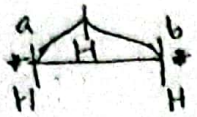
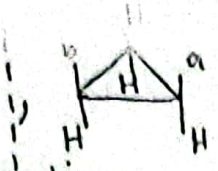
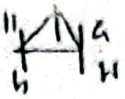
This theory only accounts for small rings but not for larger rings.



Stereoisomers of cyclopropane.

1.  ⇒ 1 chiral centre ⇒ 2^1 = 2 stereoisomers.

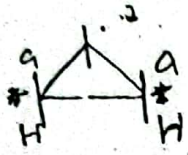
+ its mirror image. 

2.   and  

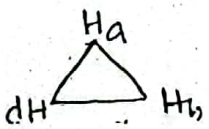
(enantiomers) (diastereomers)

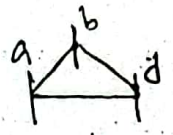
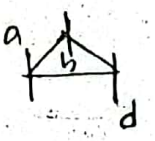
2 chiral centre ⇒ 2^2 = 4 stereoisomers.

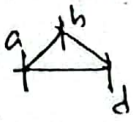
∴ 2 pair enantiomer.
1 pair geometrical isomer.

3. 

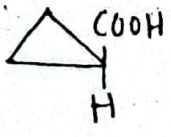
2 chiral centre ⇒ 4 stereoisomer, but there is a vertical plane of symmetry and one meso form. Thus, gets only two isomers.

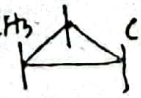
4. 

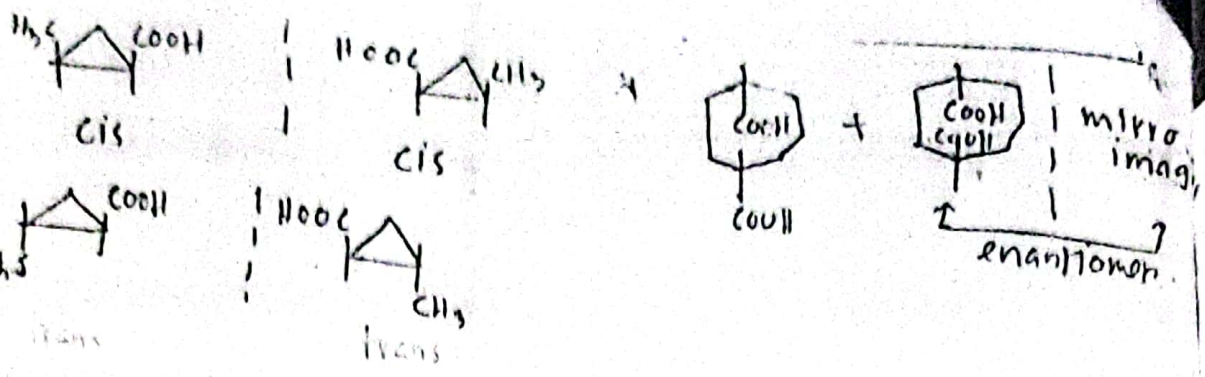
 + mirror image  + mirror image

3 chiral centre ⇒ 2^3 = 8 isomers.  + mirror image  + mirror image.

∴ 2 pair geometric isomers + 4 pair enantiomers.

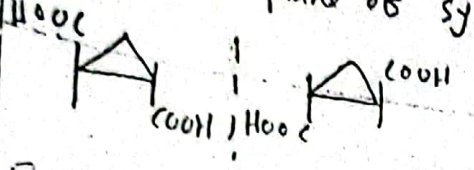
5.  ⇒ Mono substituted cyclopropane ⇒ plane of symmetry present ⇒ No isomer.

6.  ⇒ 2 chiral centre ⇒ 2^2 = 4 isomers.
1 pair geometric isomer + 2 pairs enantiomers.



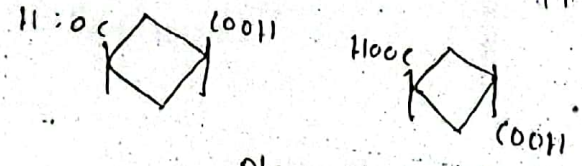


⇒ 3 isomer, since vertical plane of symmetry present.

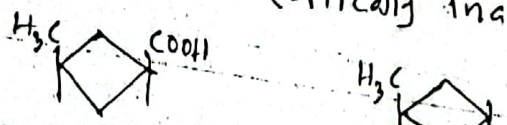


(ii) Isomers of cyclobutane :—

There is a general rule that substituents in 1 and $(1+n/2)$ position leads to plane of symmetry in the case of cyclobutane. 1, 3 $(1+4/2)$ position and also applicable for all even numbered ring.



Plane of symmetry present. (optically inactive)

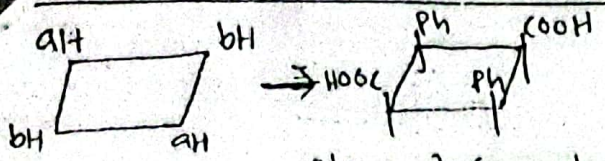


Cis-meso (vertical plane of symmetry present)

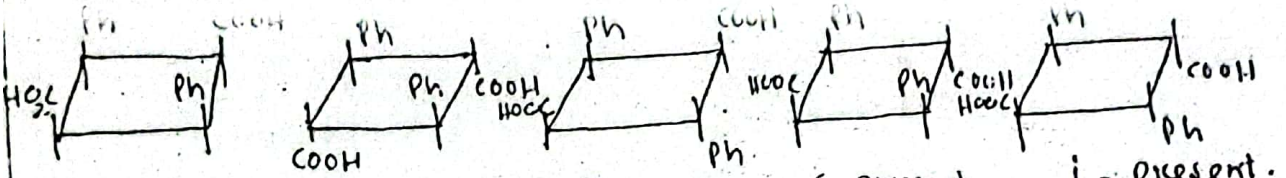
trans-meso.

Geometrical isomer.

(optically inactive)



Plane of symmetry
optically inactive
only geometrical isomers.



ζ-present ζ-present ζ-present ζ-present i-present.

(All forms are meso)

iii) Cyclobutane (small ring):-

Cyclobutane is a 4 point system and may be either planer and puckered.

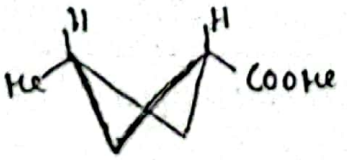
If planner then posses train due to eclipsing of adjacent hydrogen atom and to bond opposition.

But, if it is puckered, the eclipsing and bond opposition strain would be deminished, but angle strain would be increased.



-Planer

Stric interaction of adjacent H-atom.



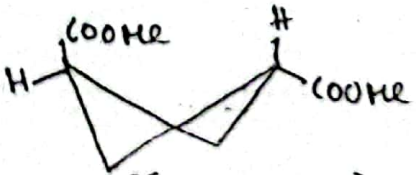
(cis)



(trans)

(Puckered) (wing shape)

Cis is more stable but alws cis is not prominent. As like the following trans structure is more favourable than cis.

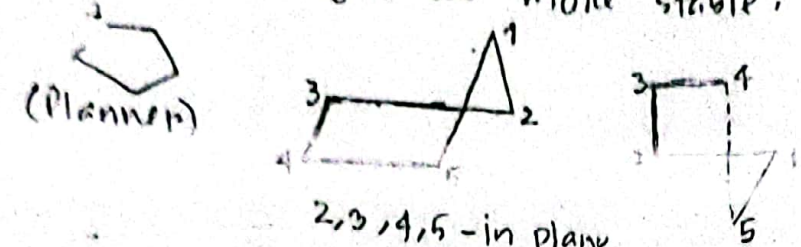


trans (more stable)

(dD/dt)

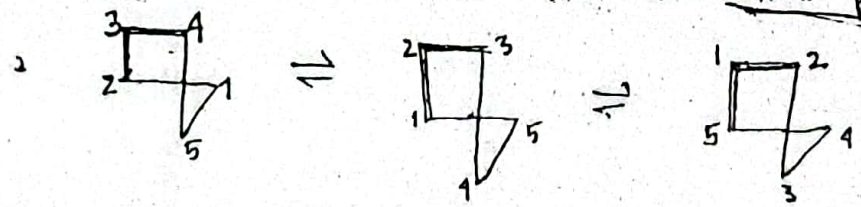
(b) Cyclopentane : (common ring)

Cyclopentane could be planner or puckered and the puckered form might be more stable.



Puckered.

The puckering is not fixed and rotates round the ring, i.e. no particular C-atom is not always out of plane but each one takes this conformation. These effect is called pseudorotation.



Continuous Oscillation (pseudorotation)

Pseudo rotation occurs in cyclopentane bcoz the energy barrier is very low, but when the ring carries one or more substituents, the energy barriers may increase and be enough to inhibit pseudorotation.

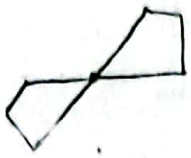
(i) Cycloheptane (common ring):-

Two stable conformation of cycloheptane is like following-



twist chair

more favourable

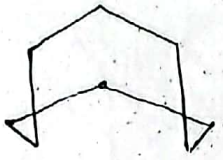


twist boat

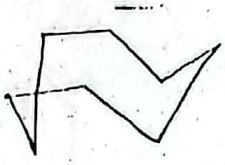
(ii) Cyclooctane (Large ring) → Medium



extended crown form.

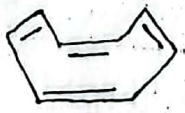


Saddle form.



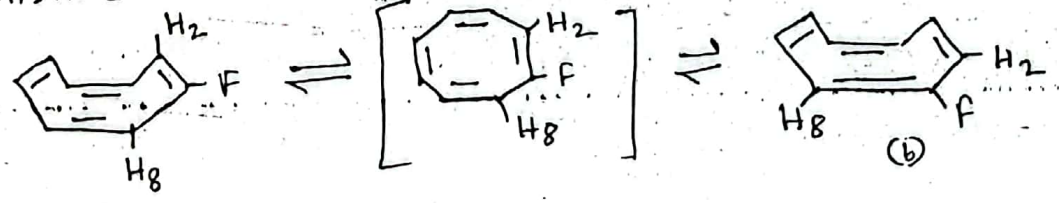
Boat-chair form

Cycloactatriene exist in tube form:



(Cycloactatriene)

The continuous bond fluctuation can be observed from this type of compound. Due to fluctuation, the following equilibrium established.



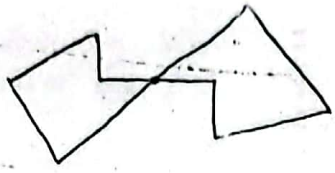
Now, at room temperature, NMR gives a wide signal. But when temperature decreased (-65°), we get two signal

(dD/dt)

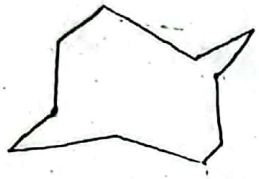
(doublet) due to coupling of H_2 with f (in fig-a) and H_8 f (in fig-b), since interconversion rate is very slow/stopped.

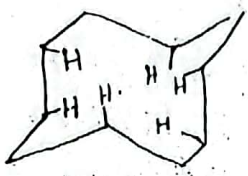
This phenomenon of bond fluctuation is called fluxionalism.

(ii) Cyclodecane :



(Puckered)





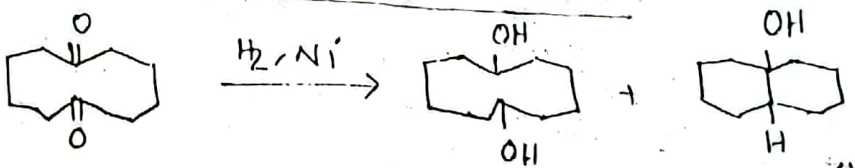
trans-cyclodecane

The H atom in trans-cyclodecane are sterically very close to one another result an extra interaction and designed as transannular interaction.

Due to transannular interaction, we get some unus behaviour of cyclodecane ring which is called transannular effect.

There are several trans annular effect.

[1] Transannular ring closer :-

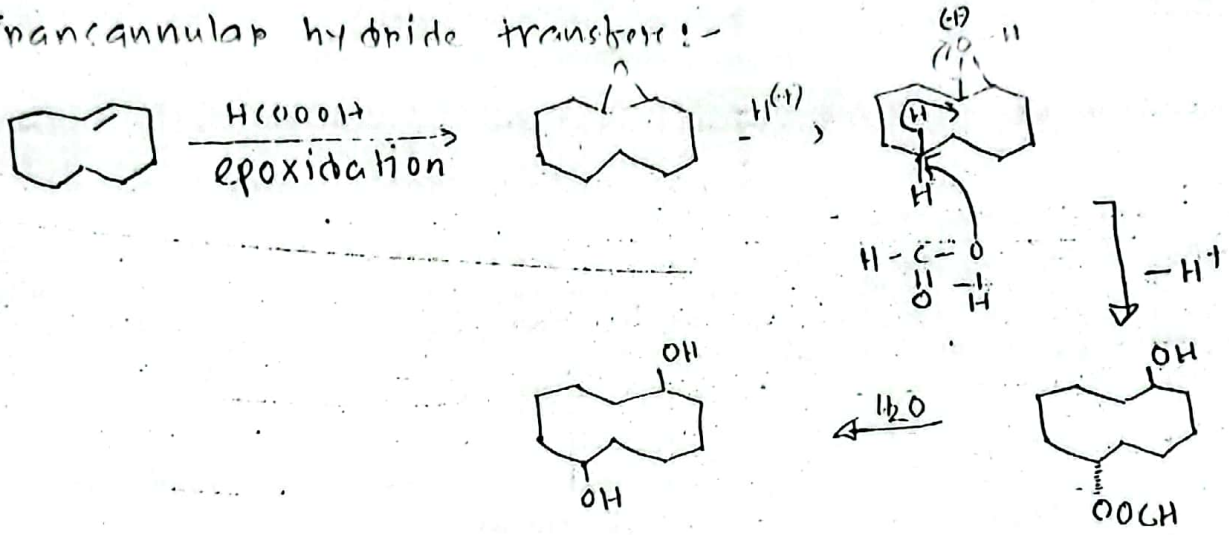


(Major product)

Side product (due to transannul effect)

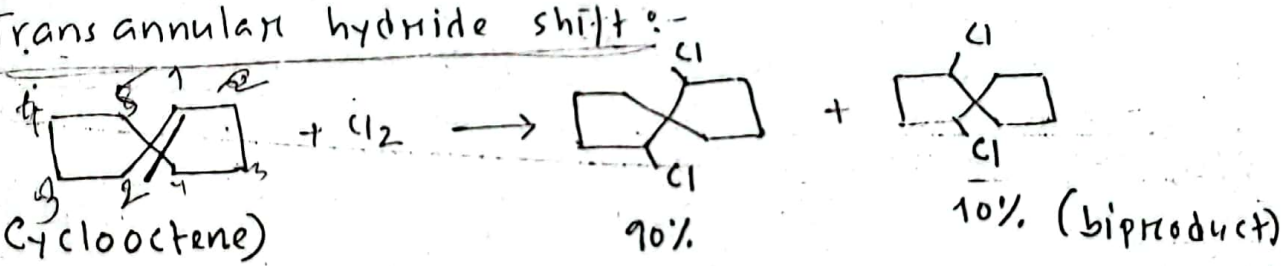
Due to transannular effect, the H is very close to the hydroxyl group, thus easily removed as water resulting ring closer.

2] Transannular hydride transfer :-



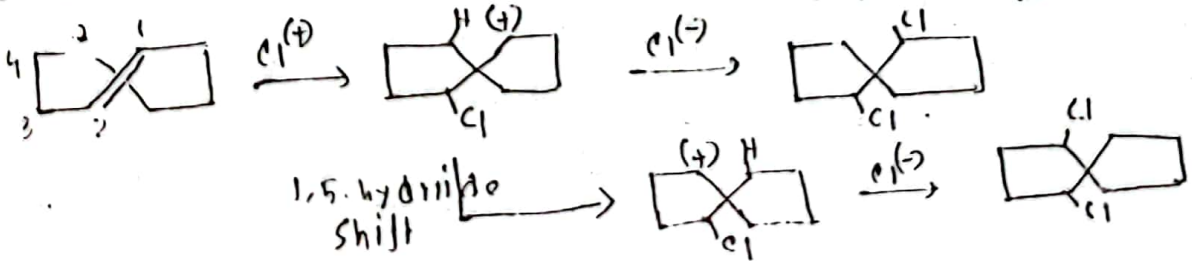
The H approaches very closer due to transannular effect and thats why H^- transfered very easily and results this type of compound.

3] Transannular hydride shift :-



(Cyclooctene) 90% 10% (biproduct)

This biproduct is formed for transannular hydride shift since the 5-positioned H atom is stereochemically closer to 1 position C-atom where, the carbocation arised.



iii Very large rings :-

In the large rings, the bond strain, angle strains are almost negligible and they behaves like open chain.



cyclo dodecane
(Strainless)

iv Cyclohexane :-

It is the most common ring; the most common conformations of cyclohexane are,



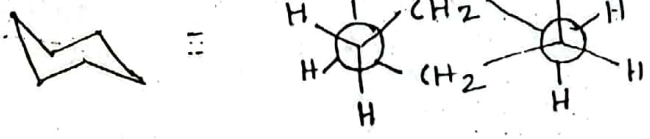
chair form



Boat form.

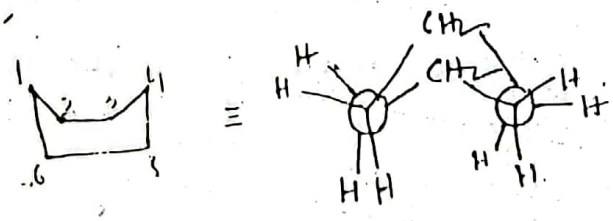
On the basis of tetrahedral theory, the above two conformation of cyclohexane are highly favourable, no one whose are planner.

The chair form is rigid i.e. resists distortion and wh it is transferred into boat form, there are some angular deformation is necessary.



1. No angular stra
2. most stable stag conformation.

But,



3. No flag-pole interaction.

1. High steric interaction.
2. Angular strain present.
3. Flag-pole strain present.

The chair and boat forms are both free from angle strain, but because of different steric strain and bond position strain, the two forms differs in energy content.

In the chair conformation, all the C-H bonds on adjacent carbons are in the skew position; on the other hand, in the boat conformation, there are four skew interactions. At the same time there will also be some bond opposition strain for these two pair of eclipsed bonds and also steric repulsion between two flag-pole.

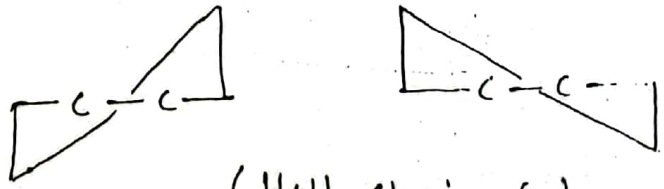
However, the boat conformation is flexible and can be destroyed readily in many shapes and in these, the hydrogen eclipsing and flag-pole interaction are reduced.



Twist boat

The twist boat conformation is 6.7 KJ/mole less energy than the classical boat conformation.

Again, Half chair conformation have the highest strain i.e. Possess less stability.

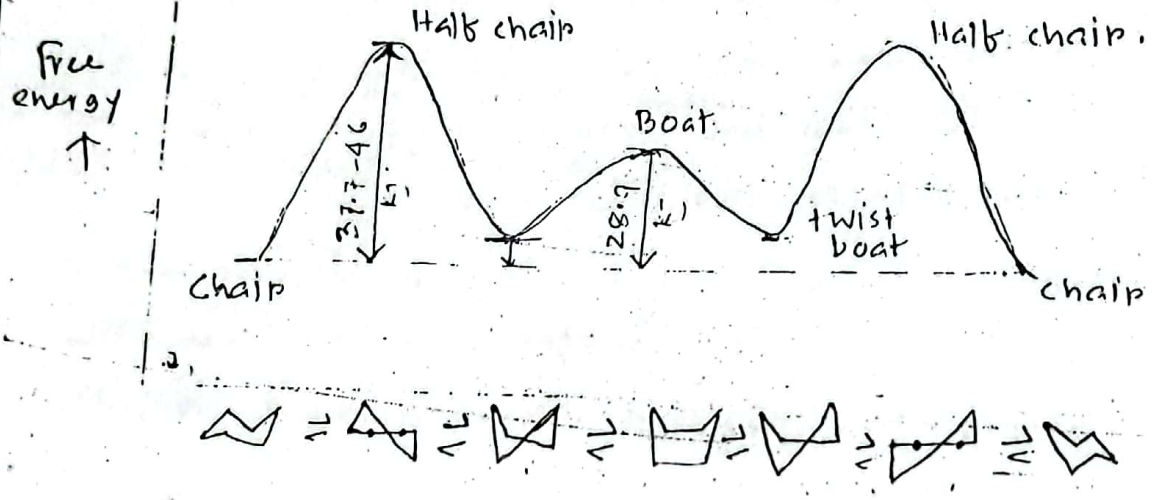


(Half chain conformation)

The energy barriers between the two chair conformation low that why they are readily interconverted, even at room temperature.



This interconversion occurs through several intermediate conformation like following –

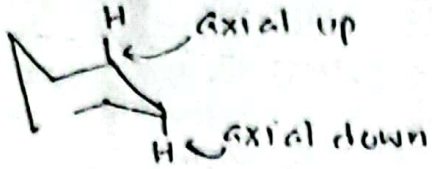


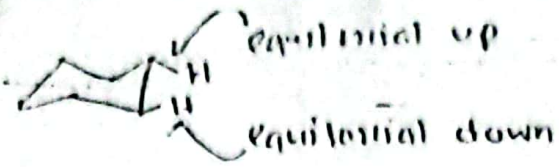
From the energy diagram, we see -

From	Energy more than chair.
Half chair →	37.7 - 46 kj
twist Boat →	22.2 kj
Boat →	28.9 kj.

The conversion barrier of two chair form is about 10 Kcal/mole, thats why, at room temperature, they exhibi rapid equilibrium.

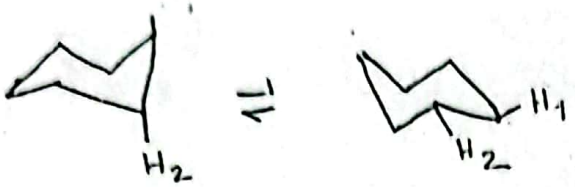
Now, there is an another factor, axial and equitorial

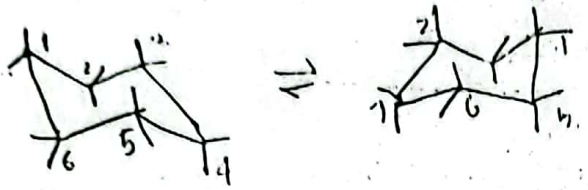




During conversion, axial up → equitorial up

axial down → equitorial down





(1) Mono substituted cycloalkane system :—



;



(more favourable)

⇒ achiral (optically inactive)

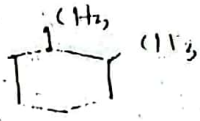
⇒ plane of symmetry present

Ex.

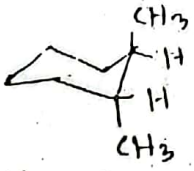


plane of symmetry

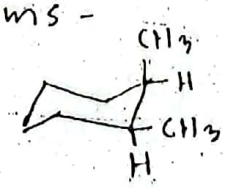
(2) 1,2 - dimethyl cyclohexane.



It can have two forms -



(trans)



(cis)

Since, No symmetry Present.

⇒ No rotational axis.

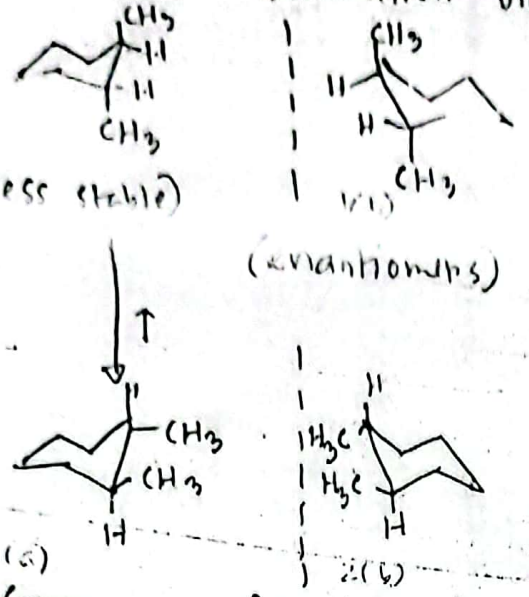
⇒ No alternating axis.

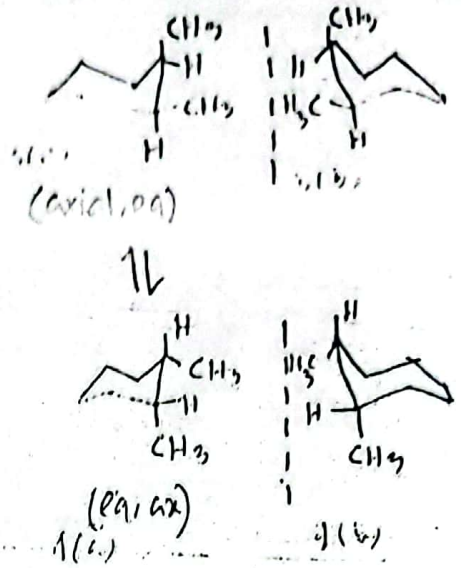
⇒ No point of inversion.

⇒ No plane of symmetry.

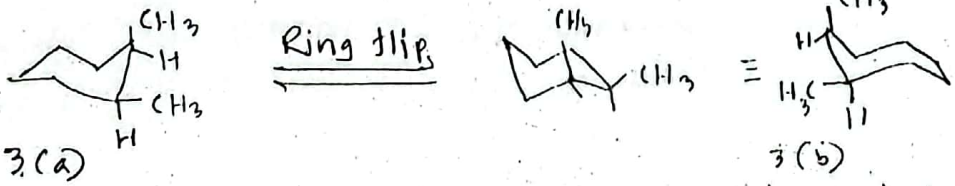
So, molecule will be achiral and will be optically active.

Consider, trans-Conformation first- consider the cis-conformation



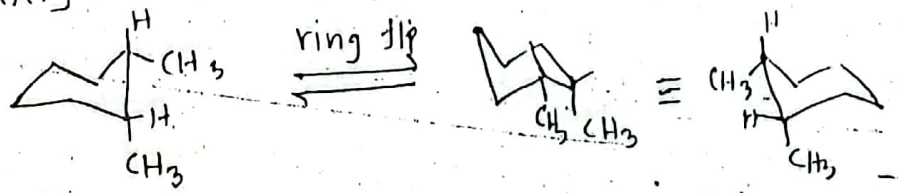


Now, we know ring flips readily due to low energy barrier. Thats why we will have to consider the structure - results often the ring flipping.



Ring flip में
axial → eq
equitorial →

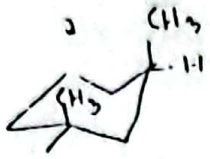
Similarly,



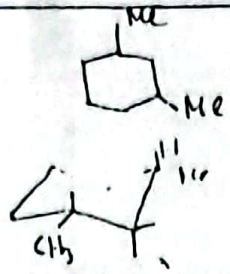
Thus, though the compound is optically active chiral but will not show optical activity and exist as racemic mixture. In cis- isomer and two (1) enantiomer from trans. So, total three isomers obtained instead of four.

1.3 - dimethyl cyclohexane:

cis :-

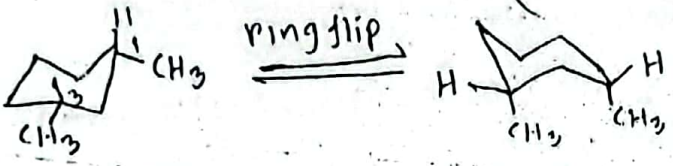


1(a)
(less stable)

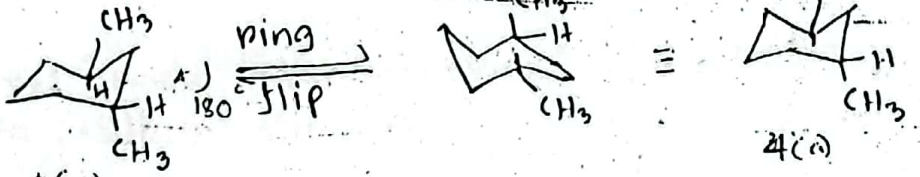


(meso compound)
(optically inactive)

2(a)
(more stable)



Trans :-

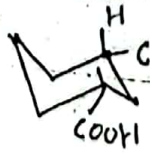


3(a)

4(a)

Ring flips generate the similar structure. Among the all 2(a) is most stable since both two big groups are in equitorial position. Here, racimization not occures in trans form and the compnd is chiral and optically active.

ii) cis - 1,3 - cyclohexane dicarboxylic acid :-

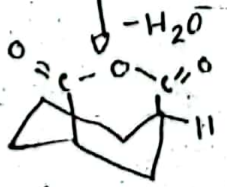


but exist as



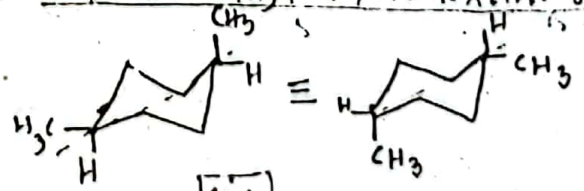
since forms anhydride

(more stable)
(eq, eq)

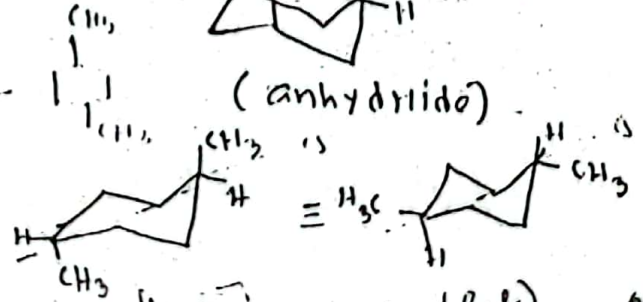


(anhydride)

iii) 1,4 - dimethyl cyclohexane :-



cis

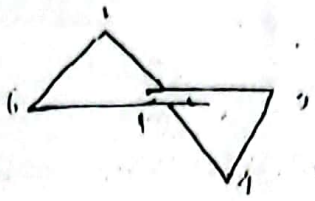


trans

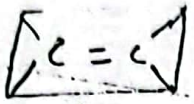
trans (e,e) more stable.

=) All are achiral.
=) δ- present and thus optically inactive.

iii) Cyclohexane: [1]

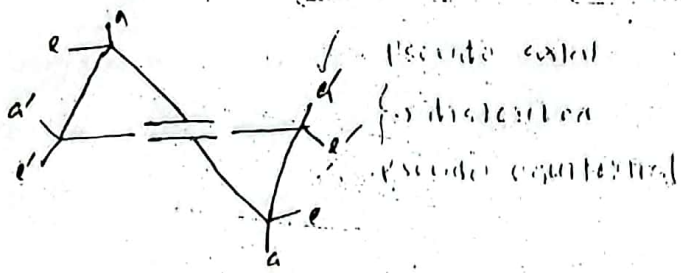
It will exist in half chair form. 

1, 2, 3, 6 carbons will lie almost co-planar.



since, sp^2 hybridised,
C_2 will pull C_3 to be coplannered;
C_1 " " C_6 " ;

In cycloalkene system, A-carbon remain almost coplaner and thats why the axial and equitorial positions are distorted from originate, called pseudo axial and pseudo equitorial



iv) 2-Alkyl Ketone effect:-
3-Alkyl Ketone effect:-

A Carbonyl (>C=O) mostly placed equitorial position when less interaction arise.

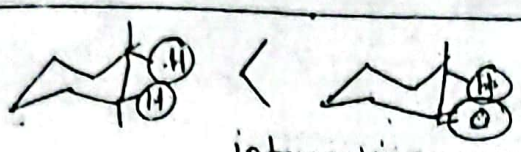


(more stable)
(less interaction)



(more interaction)

Now, if a substituent remain in 2-equitorial position, then due to O atom, interaction will be stronger than normal cyclohexane. This phenomenon is called 2 Alkyl Ketone effect.



; 2- Alkyl Ketone effect.

interaction.

On the other hand, due to C. C., one hydrogen less than original cyclohexane, results less axial interaction.



; 3- alkyl Ketone effect.

Similarly, the different behaviour arised due to less axial interaction in cyclohexanone is called 3-alkyl Ketone effect.

The substituent in axial/equitorial effect the C=O bond, can be visualised by UV/IR spectra :-



IR ⇒ Normal ± 20 cm^{-1}

UV ⇒ shorter λmax due to bond strength.

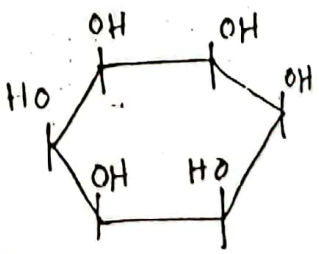
Normal

longer λmax due to successive overlap of Π-electron.

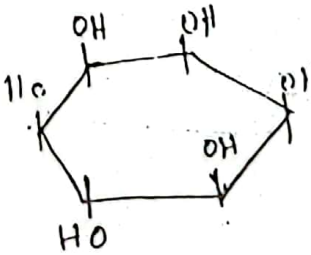
Inositol system :-

2004
2006

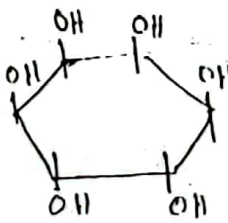
The hexa substituted cyclohexane system is called inositol system, $C_6H_6X_6$ which have 8 cis-trans isomers. Of which 7 are meso and only one is optically active.



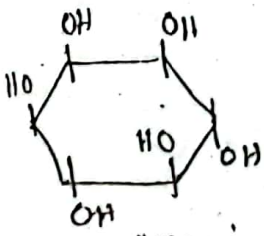
cis-inositol (i)



Epiinositol (6)



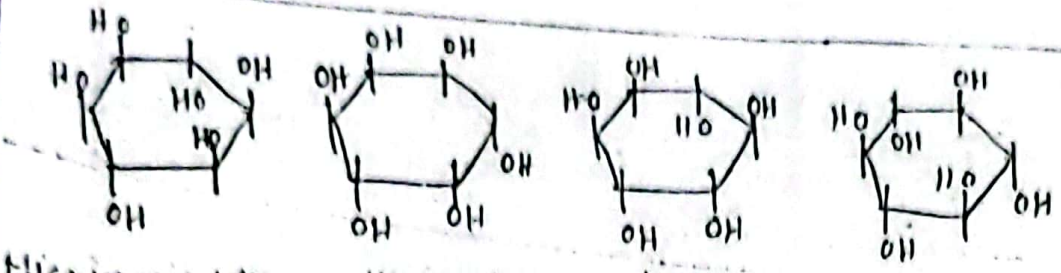
Alloinositol (5)



meso inositol (3)

ovi ox

myo (4,6) cino (1,5) muco (1,4) allo (1,2).



Alloinositol (✓) Neoinositol (✓) Myo-inositol (✓) Scyllo-inositol (✓)

Effect of conformation on the course and rate of reaction

The conformational variety may differ in the product of same reaction. As we can consider for nucleophilic substitution reaction.

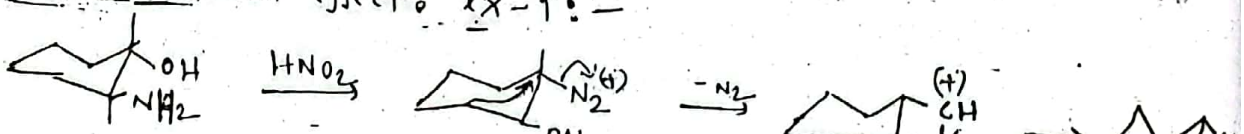


easier (less hindered) Reactivity difficult (most hindered site)

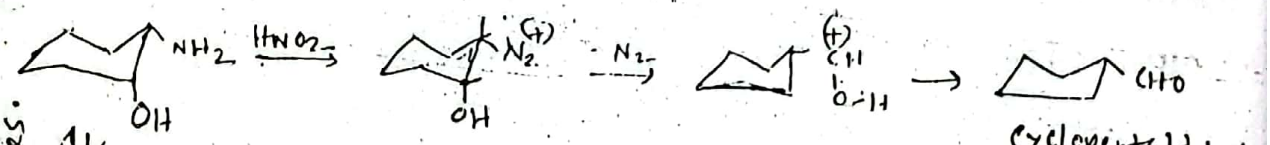
But, this consideration is not always prominent. Reactant may be fastest from equitorial position also.

Conformational effect: ex-1:-

2005 Explain

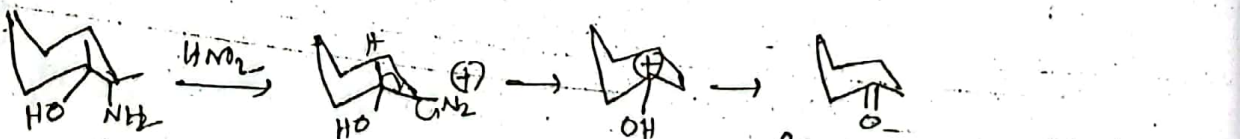


trans-2-amino cyclohexanol



Intermediate forms and departure of N_2 is antiperiplanar with C_2–C_3 bond. Thus C_2–C_3 migrates.

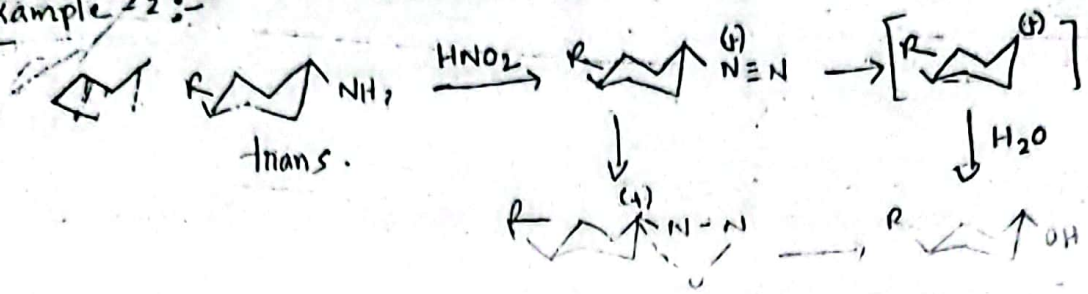
1V



cis- (Cyclopentanone) (Cyclohexanone)

aldehyde and ketone obtained where a coplanar H is present.

Example-2 :-



The OH added in equitorial position since most stable conformation.

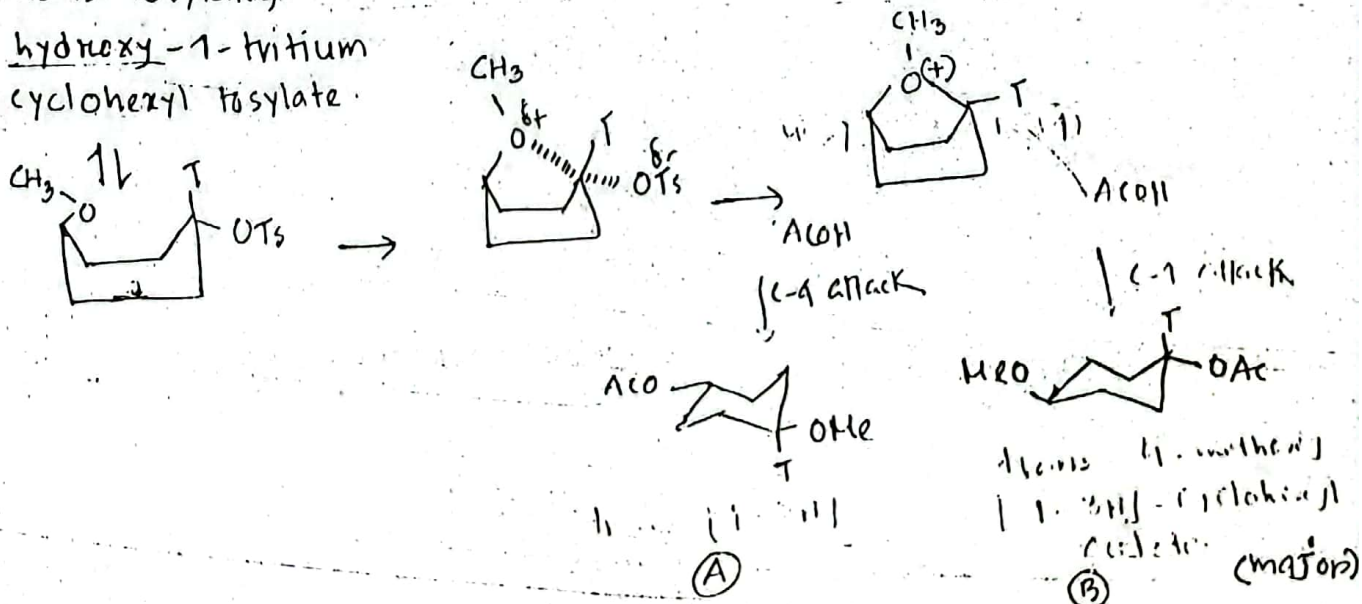
But the cis conformation gives different product.



Example-3 :

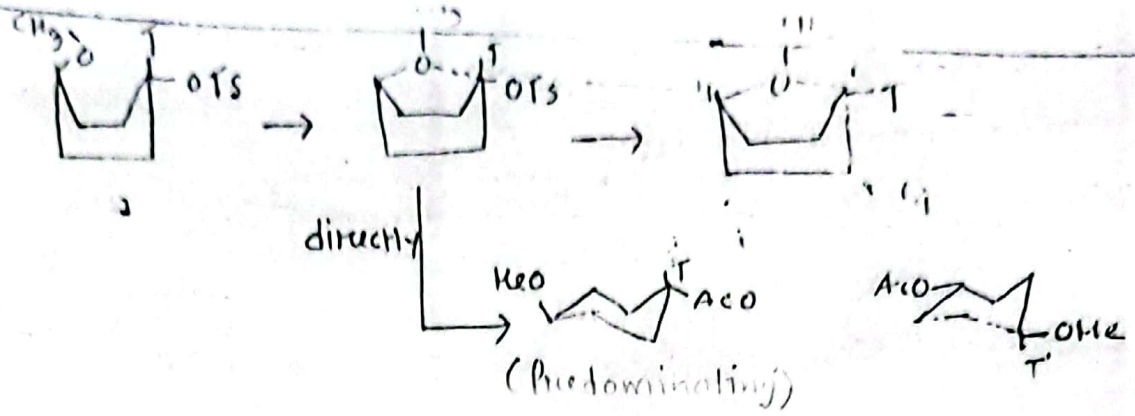


4-hydroxy-1-tritium cyclohexyl tosylate.

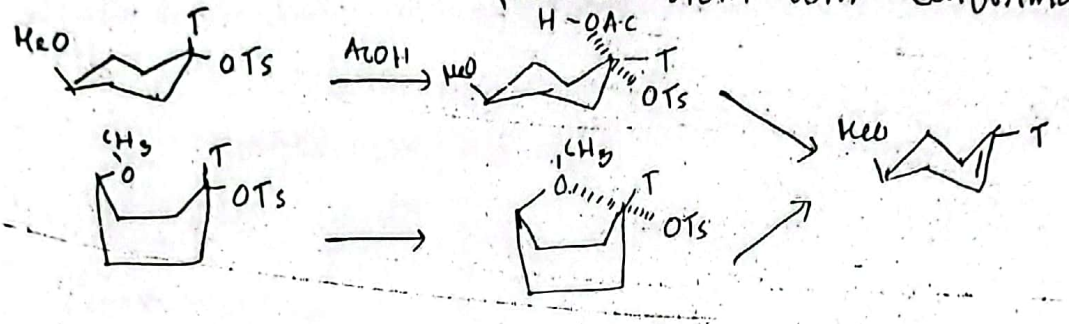


(A) and (B) should be produced in equal amount, since AcOH have equal probability and opportunity to attack at C1 and C4.

But, (B) is predominating product, since addition to normal way, it can be further produced from the transition st directly.



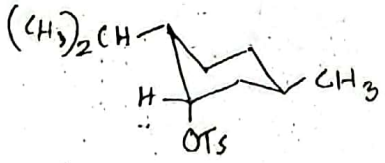
we got a common product from both conformation also—



Thus, chair conformation gives only one product, but both conformation gives two, one is predominating.

Example-4 :-

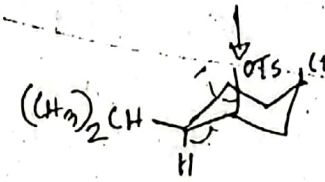
2006 Neomenthyl tosylate gives 170 times faster solvolysis reaction than corresponding menthyl tosylate.



(Neomenthyl tosylate)

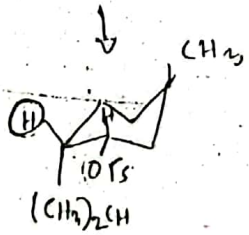


(Menthyl tosylate) (more stable)



(Easy removal of H)

OTs and H are in antiperriplanar position thus elimination is easy and gives reaction very fast.



OTs and H are not in anti position, so, removal is not easier.

Curtin - Hammett principle.

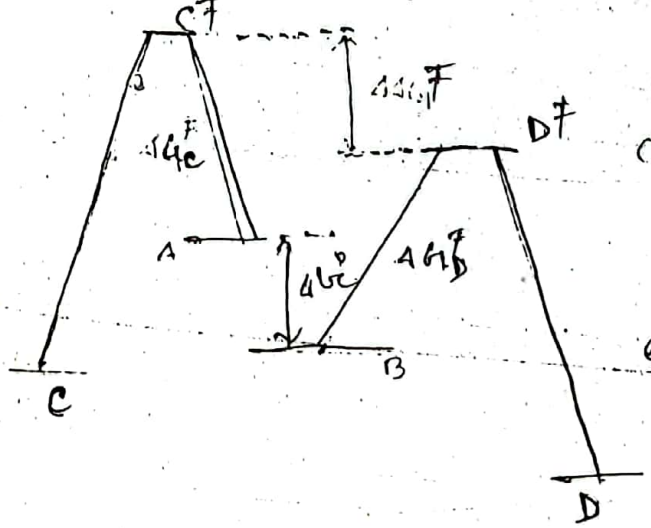
2006

The principle states that, in a chemical reaction that yields one product from one conformer and a different product from another conformer (and these conformers are readily interconvertible relative to the rate of product formation, where the products are not interconvertible), then the product composition is not solely dependent on the relative proportion of the conformers in the substrate. It is controlled by the difference in the standard Gibbs energies of the representative transition state.

Let, A and B are two conformers existing in equilibrium.

$$C \xleftarrow{K_C} A \underset{K_B}{\overset{K_A}{\rightleftharpoons}} B \xrightarrow{K_D} D$$

The energy level diagram can be designed as like following -



Now, there may be three cases -

Case-1 :- $K_A, K_B >> K_C, K_D$ then, equilibrium is attained betw the conformers. As a result,

eq. constant, $K = \frac{[D]}{-[A]} = \frac{K_A}{K_B}$

Case-2 :- $K_C, K_D >> K_A, K_B$

This is called kinetic quenching since, no equilibrium is attained betw the conformers. Here, $[C]/[D]$ reflects $[A]/[B]$

Case-3 : $K_A, K_B = K_C, K_D$, Here, no equilibrium is mentioned during the reaction and both of the two principle are non-valid.

2004

// NOW, the overall rate of the reaction,

$$\frac{d[A+B]}{dt} = K_{WH}[A+B] \quad \text{---- (1)}$$

K_{WH} = Winstein-Holness rate co

Again, $\frac{d[A]}{dt} + \frac{d[B]}{dt} = K_C[A] + K_D[B]$ ... (2)

For case-1: $K = [B]/[A] \Rightarrow [B] = [A]K$ (3)

equating eqn (1) and (2) $\Rightarrow$

$$K_C[A] + K_D[B] = K_{WH}[A+B]$$

$$\Rightarrow K_C[A] + K_D K[A] = K_{WH}\{[A] + [A]K\}$$

$$\Rightarrow K_C + K_D K = K_{WH}\{1+K\}$$

$$\Rightarrow K_{WH} = \frac{K_C + K_D K}{1+K} \quad \text{---- (4)}$$

This is called Winstein-Holness eqn.

Again, let n_A and n_B the mole fraction of A and B.

So, $K = \frac{[B]}{[A]} = \frac{n_B}{n_A}$ ---- (5)

eqn (4) $\Rightarrow$

$$K_{WH} = \frac{(K_C + K_D\, n_B/n_A)}{(1 + n_B/n_A)} = \frac{n_A K_C + n_B K_D}{n_A + n_B} \quad [\because n_A + n_B = 1]$$

$$\Rightarrow K_{WH} = n_A K_C + n_B K_D \quad \text{---- (6)}$$

Expanding this eqn for ith no. of conformers,

$$K_{WH} = \sum_i n_i k_i$$

Thats why, the observed rate constant for a conformationally heterogeneous system, under condition of fast conformational interchange is the weight average of all the individual rate constant for the contributing conformer.

We can design individual rate -

$$\frac{d[C]}{dt} = K_C[A] \quad \Big\} \text{---(8)}$$
$$\frac{dD}{dt} = K_D[A]$$

And eq. constant, $K = \frac{[D]}{[A]}$ ---(9)

So, product ratio, $\frac{dC/dt}{dD/dt} = \frac{K_D[A]}{K_C[A]} = \frac{K_D}{K_C} K$ -

$$\Rightarrow \frac{dC/dt}{dD/dt} = \frac{K_D}{K_C} K \text{ --- (10)}$$

From Boltzmann equation, $K_D = K \frac{KT}{h} e^{\frac{-\Delta G_D^{\neq}}{RT}}$ ---- 11(a)

$$K_C = K \frac{KT}{h} e^{\frac{-\Delta G_C^{\neq}}{RT}} \text{ ---- 11(b)}$$

We can rationalize equilibrium constant with Gibbs free energy

$$\Delta G^\circ = -RT \ln K$$
$$\Rightarrow K = e^{-\Delta G^\circ / RT} \text{ ---- (12)}$$

Now, employing the value from 11(a), (b) and (12) into eq^n (10)

$$\frac{dC/dt}{dD/dt} = \frac{e^{\frac{-\Delta G_D^{\neq}}{RT}}}{e^{\frac{-\Delta G_C^{\neq}}{RT}}} \cdot e^{-\Delta G^\circ / RT}$$

$$\Rightarrow \frac{dC/dt}{dD/dt} = \frac{e^{\frac{-\Delta G_D^{\neq} - \Delta G^\circ}{RT}}}{e^{-\Delta G_C^{\neq}/RT}} = e^{\frac{-\Delta G_D^{\neq} - \Delta G^\circ + \Delta G_C^{\neq}}{RT}} \text{ ---- (13)}$$

Now, from figure, we get.

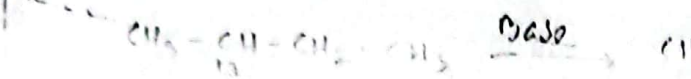
The difference of Gibbs free energy betw^n two transition state

$$\Delta\Delta G^{\neq} = \Delta G_D^{\neq} - \Delta G_C^{\neq} + \Delta G^\circ$$
$$\Rightarrow -\Delta\Delta G^{\neq} = -\Delta G_D^{\neq} - \Delta G^\circ + \Delta G_C^{\neq}$$

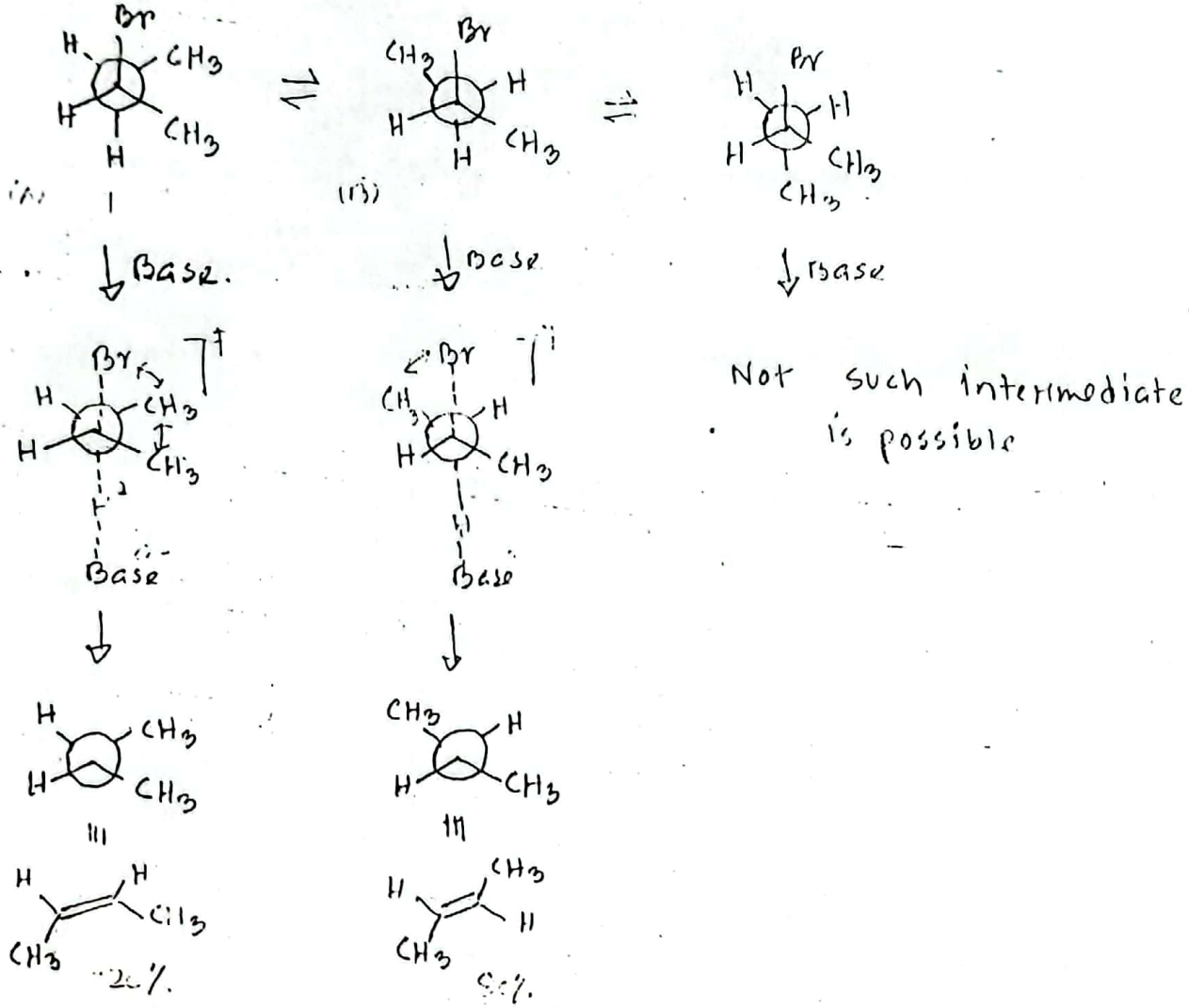
So, Product ratio; $\left(\frac{dC/dt}{dD/dt}\right) = e^{\frac{-\Delta\Delta G^{\neq}}{RT}}$ -

Thus, product ratio depend on the difference of gibbs free energy of representative transition states, not on the ratio of the conformer.

As example:



Now, 2-Bromobutane has different conformations and product may be cis/trans-



The transition state of (A) conformer is more hindrance. thats why energy barriers is high, on the other hand, corresponding transition states of (B) is less hindered, as a result, energy barriers is relatively lower.

Since energy barriers is low, corresponding Prod